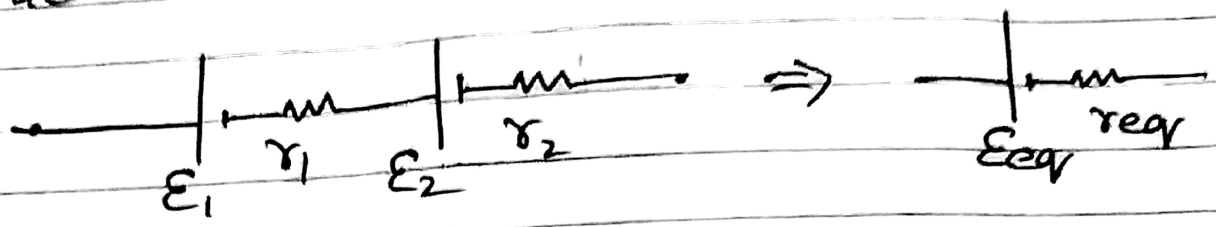
# Combination of Cells

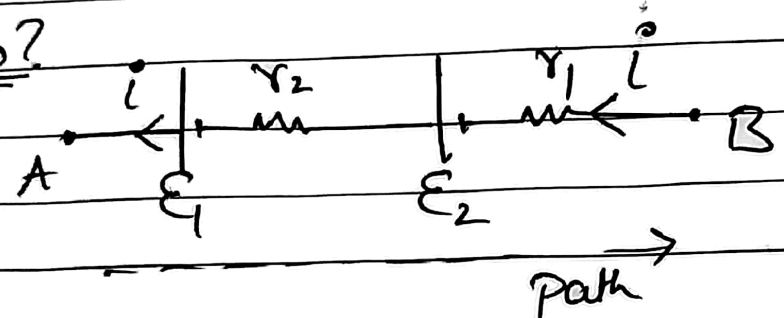
## (1) Series

(i)




$$\mathcal{E}_{eq} = \mathcal{E}_1 + \mathcal{E}_2 \qquad r_{eq} = r_1 + r_2$$

How?




$$V_A - \mathcal{E}_1 + i r_2 - \mathcal{E}_2 + i r_1 = V_B$$

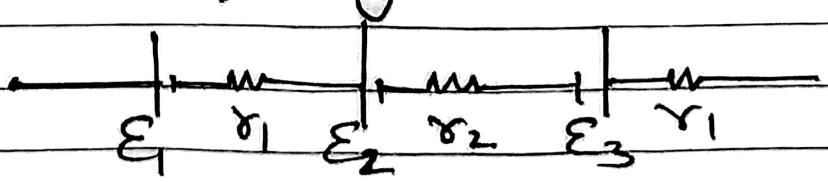
$$V_A - V_B = (\mathcal{E}_1 + \mathcal{E}_2) - i(r_1 + r_2)$$

$$V = (\mathcal{E}_1 + \mathcal{E}_2) - i(r_1 + r_2)$$

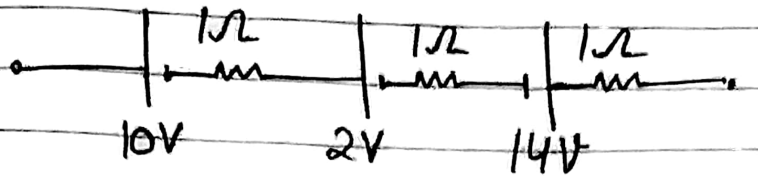
Comparing with $V = \mathcal{E} - ir$

$$\mathcal{E} = \mathcal{E}_1 + \mathcal{E}_2 \qquad r = r_1 + r_2$$

(ii) If polarity of any cell is reversed



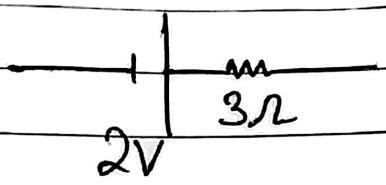

$$\mathcal{E}_{eq} = \mathcal{E}_1 + \mathcal{E}_2 - \mathcal{E}_3 \qquad r_{eq} = r_1 + r_2 + r_3$$

Q1) 

Find eq $\mathcal{E}$ & $r$

$\mathcal{E}_{eq} = 10+2-14 = -2V$ (opp to 10V polarity)

$r_{eq} = 1+1+1 = 3\Omega$




if 'n' cells each of EMF $\mathcal{E}$ & internal resistance 'r' are connected in series in same polarity order.

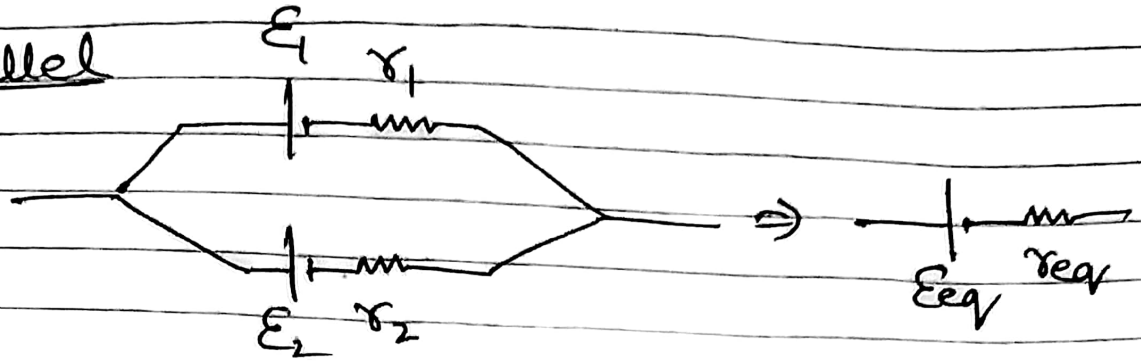
$$\mathcal{E}_{eq} = n\mathcal{E} \qquad r_{eq} = nr$$

Q2) If 'n' cells are connected in series with -ve terminal of one connected to +ve of other & so on. If polarity of 'm' cells are reversed. Find the Equivalent EMF. (EMF of each cell is $\mathcal{E}$)

$$\mathcal{E}_{eq} = n\mathcal{E} - m\mathcal{E} - m\mathcal{E}$$

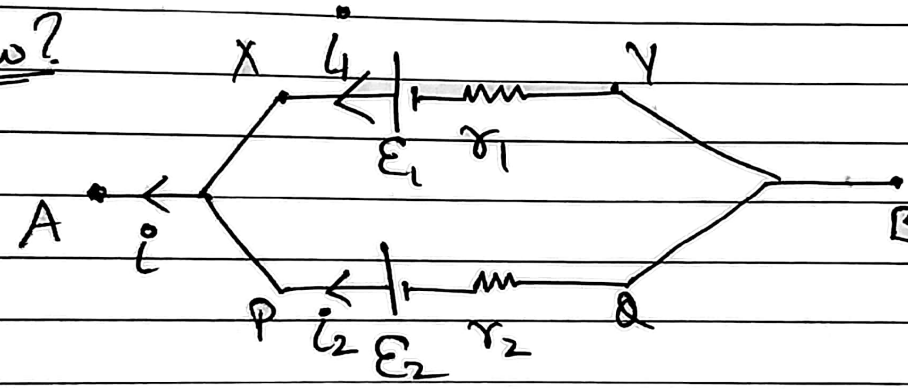
$$= (n-2m)\mathcal{E}$$

## ② Parallel




$$\varepsilon_{eq} = \frac{\frac{\varepsilon_1}{r_1} + \frac{\varepsilon_2}{r_2}}{\frac{1}{r_1} + \frac{1}{r_2}} \qquad r_{eq} = \frac{r_1 r_2}{r_1 + r_2} \text{ or } \frac{1}{r_{eq}} = \frac{1}{r_1} + \frac{1}{r_2}$$

**How?**




$$i = i_1 + i_2$$

**Path AXYB**

$$V_A - \varepsilon_1 + i_1 r_1 = V_B$$

$$V_A - V_B = \varepsilon_1 - i_1 r_1$$

Let $V = \varepsilon_1 - i_1 r_1$

$$i_1 = \frac{\varepsilon_1 - V}{r_1}$$

Similarly from path APQB

$$i_2 = \frac{\varepsilon_2 - V}{r_2}$$

$$i = i_1 + i_2$$

$$i = \frac{\mathcal{E}_1 - V}{r_1} + \frac{\mathcal{E}_2 - V}{r_2}$$

$$i = \frac{\mathcal{E}_1}{r_1} + \frac{\mathcal{E}_2}{r_2} - V\left(\frac{1}{r_1} + \frac{1}{r_2}\right)$$

$$V\left(\frac{1}{r_1} + \frac{1}{r_2}\right) = \frac{\mathcal{E}_1}{r_1} + \frac{\mathcal{E}_2}{r_2} - i$$

$$V = \frac{\frac{\mathcal{E}_1}{r_1} + \frac{\mathcal{E}_2}{r_2}}{\frac{1}{r_1} + \frac{1}{r_2}} - i\left(\frac{1}{\frac{1}{r_1} + \frac{1}{r_2}}\right)$$

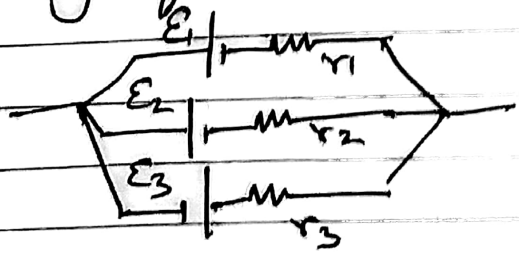
Comparing with $V = \mathcal{E} - ir$

$$\mathcal{E}_{eq} = \frac{\frac{\mathcal{E}_1}{r_1} + \frac{\mathcal{E}_2}{r_2}}{\frac{1}{r_1} + \frac{1}{r_2}} \qquad r_{eq} = \frac{1}{\frac{1}{r_1} + \frac{1}{r_2}}$$

or

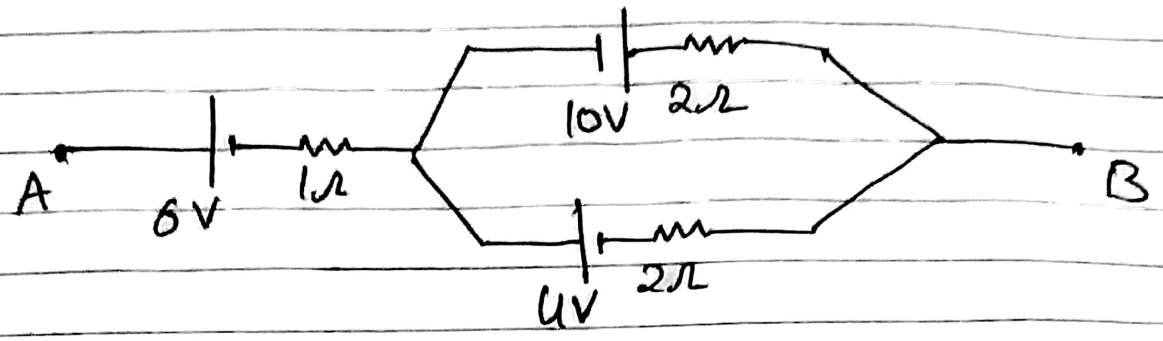
$$\frac{1}{r_{eq}} = \frac{1}{r_1} + \frac{1}{r_2}$$

(ii) If polarity of one cell is opp




$$\mathcal{E}_{eq} = \frac{\frac{\mathcal{E}_1}{r_1} + \frac{\mathcal{E}_2}{r_2} - \frac{\mathcal{E}_3}{r_3}}{\frac{1}{r_1} + \frac{1}{r_2} + \frac{1}{r_3}} \qquad \frac{1}{r_{eq}} = \frac{1}{r_1} + \frac{1}{r_2} + \frac{1}{r_3}$$

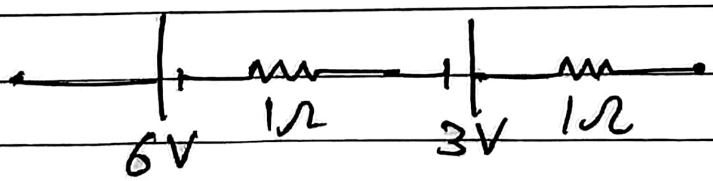
Q2) Find equivalent EMF & internal resistance.




Solution 10V & 4V in parallel

$$E_{eq} = \frac{\frac{10}{2} - \frac{4}{2}}{\frac{1}{2} + \frac{1}{2}} = 3V \quad \text{(in polarity of 10V)}$$

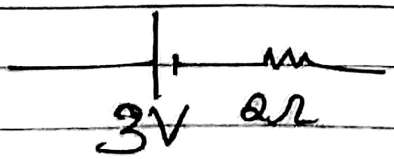
$$\frac{1}{r_{eq}} = \frac{1}{2} + \frac{1}{2} = 1 \qquad r_{eq} = 1$$

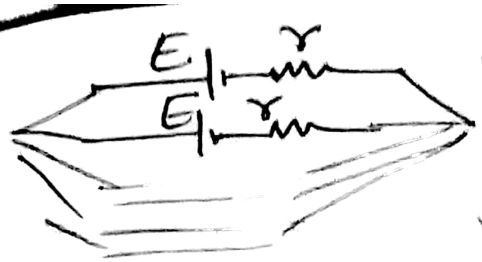



6V & 3V in series

$E_{eq} = 6 - 3 = 3V$ (in polarity of 6V)

$r_{eq} = 1 + 1 = 2\Omega$






$E_{eq} = E$

$r_{eq} = \frac{r}{n}$

if 'n' cells each of EMF E & internal resistence r are connected in parallel.

$$E_{eq} = \frac{\frac{E}{r} + \frac{E}{r} + \frac{E}{r} + \cdots \text{ n times}}{\frac{1}{r} + \frac{1}{r} + \frac{1}{r} + \cdots \text{ n times}}$$

$$E_{eq} = \frac{\frac{nE}{r}}{\frac{n}{r}} = E$$

$\boxed{E_{eq} = E}$ ?? Yes!!

same as one

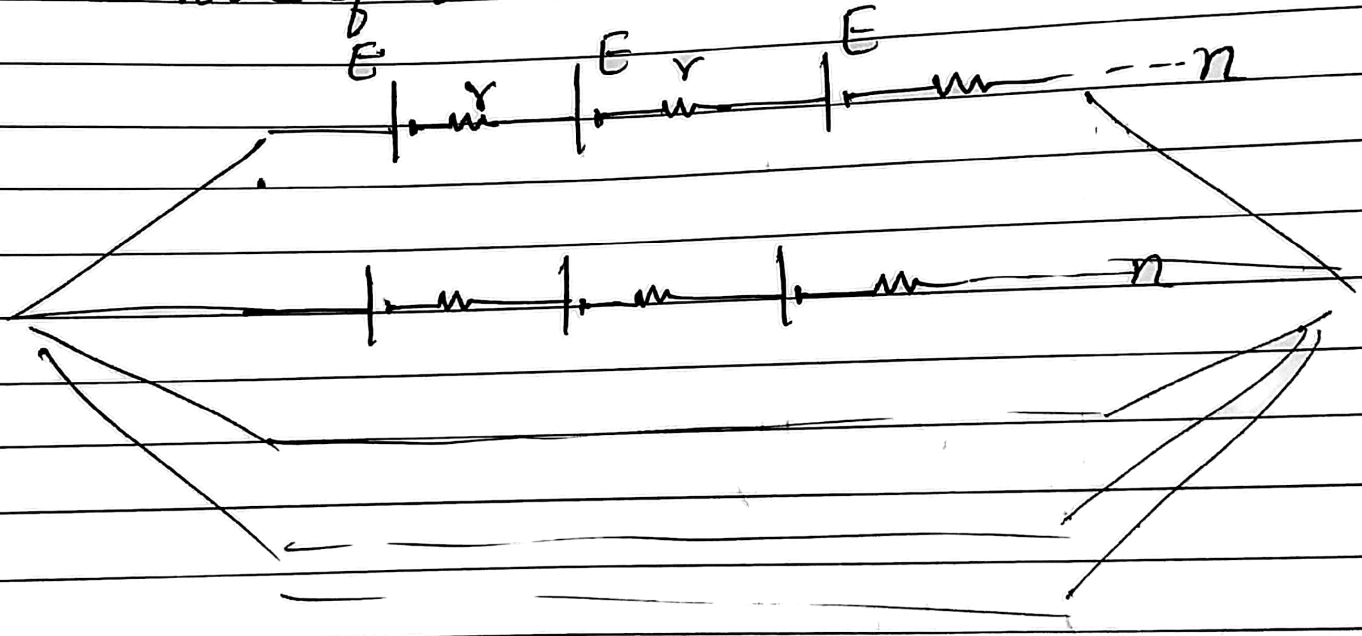
$$\frac{1}{r_{eq}} = \frac{1}{r} + \frac{1}{r} + \frac{1}{r} + \frac{1}{r} \cdots \text{ n times} = \frac{n}{r}$$

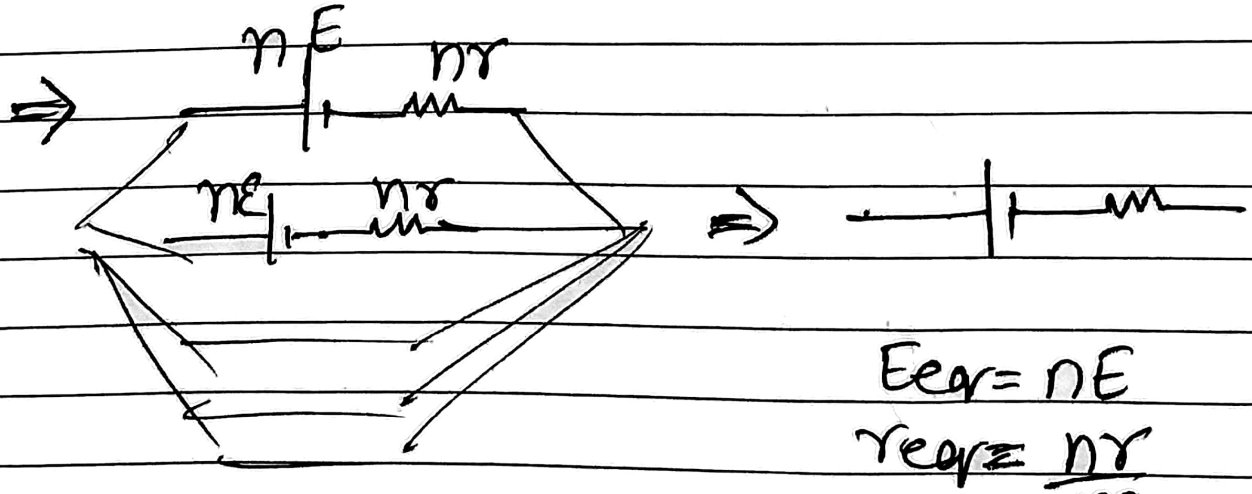
$$\boxed{r_{eq} = \frac{r}{n}}$$

# Mixed Grouping

Q) If cells of EMF 'E' are arranged in 'm' rows connected in parallel such that each row contains 'n' cells. Find (Internal resis → r of each)

i) no. of cells
ii) $E_{eq}$
iii) $r_{eq}$
iv) Current 'i' if external resistance 'R' is connected
v) condition for 'i' to be maximum
vi) Max value of i.




$E_{eq} = nE$

$r_{eq} = \frac{nr}{m}$

$$i = \frac{nE}{\frac{nr}{m} + R} = \frac{mnE}{nr + mR}$$

$i \longrightarrow max \qquad nr + mR \longrightarrow min$

$$nr + mR = \left(\sqrt{nr} - \sqrt{mR}\right)^2 + 2\sqrt{nmrR}$$

$nr + mR \longrightarrow min$

$\left(\sqrt{nr} + - \sqrt{mR}\right)^2 \longrightarrow 0 \qquad \sqrt{nr} = \sqrt{mR}$

$nr = mR$

$$\boxed{R = \frac{nr}{m}}$$

net external resistance = net internal resistance

$$i_{max} = \frac{mnE}{nr + mR} = \frac{mnE}{mR + mR} = \frac{nE}{2R}$$